**tsvc/s281 · s281.c · bare_llm · claude-haiku-4-5-20251001 · FASTER**

−1 / +21 lines of 178 · 2 parallel region(s) added · 0 of 0 candidate(s) kept · RESTRUCTURING (6 code line(s) added, 1 removed) · restructuring level 0

BEFORE

```
133
134 static real_t kernel_s281(void)
135 {
136     real_t x;
137     for (int nl = 0; nl < R; nl++) {
138         for (int i = 0; i < LEN_1D; i++) {
139             x = a[LEN_1D-i-1] + b[i] * c[i];
140             a[i] = x-(real_t)1.0;
141             b[i] = x;
142         }
143         pb_mix(nl);
144     }
145     return (real_t)0;
146 }
147
```

AFTER

```
133
134 static real_t kernel_s281(void)
135 {
136     real_t x;
137     for (int nl = 0; nl < R; nl++) {
138         /* First half: i from 0 to LEN_1D/2-1
139          * No cross-iteration dependencies. Each iteration i reads a[LEN_1D-i-
140          * which hasn't been written yet (since writes are to a[j] where j < L
141          * Private x: each thread gets its own scratch variable for the comput
142          */
143         #pragma omp parallel for private(x)
144         for (int i = 0; i < LEN_1D / 2; i++) {
145             x = a[LEN_1D-i-1] + b[i] * c[i];
146             a[i] = x-(real_t)1.0;
147             b[i] = x;
148         }
149
150         /* Implicit barrier between parallel regions ensures first half comple
151          * Second half: i from LEN_1D/2 to LEN_1D-1
152          * Each iteration i reads a[LEN_1D-i-1] which was written by iteration
153          * (LEN_1D-i-1) in the first half. No cross-iteration dependencies
154          * within the second half.
155          */
156         #pragma omp parallel for private(x)
157         for (int i = LEN_1D / 2; i < LEN_1D; i++) {
158             x = a[LEN_1D-i-1] + b[i] * c[i];
159             a[i] = x-(real_t)1.0;
160             b[i] = x;
161         }
162
163         pb_mix(nl);
164     }
165     return (real_t)0;
166 }
167
```

file

red = removed  green = added  yellow = parallel region (#pragma omp)  right strip = the whole file: green where changed, yellow where a parallel region now is